EXERCISE 7

CODE 1(*Linear Search):*

```
#include <iostream>
using namespace std;

int linearSearch(int arr[], int size, int target){
        for (int i = 0; i<size; i++){
        if(arr[i]==target){
        return i;//Return the index if found
        }
        }
        return-1;//Return -1 if not found
}
int main(){
        int arr[]={5,3,8,4,2};
        int size = sizeof(arr)/sizeof(arr[0]);
        int target;

        cout<<"Enter a number to search: ";
        cin >>target;

        int result=linearSearch(arr, size, target);
        if(result!=-1){
        cout<<"Element found at index: "<<result<<endl;
        }else{
        cout<<"Element not found."<<endl;
        }
        return 0;
        }
```

OUTPUT:

Output

```
/tmp/mHvdmovswd.o
Enter a number to search: 5
Element found at index: 0


=== Code Execution Successful ===
```

CODE 2 (*Binary Search):*

```
#include <iostream>
using namespace std;

int binarySearch(int arr[], int size, int target){
        int left =0;
        int right = size -1;

        while (left<=right){
        int mid = left +(right - left)/2;

        if (arr[mid]==target){
        return mid;
        }
        if (arr[mid]<target){
        left = mid +1;
        }else{
        right = mid -1;
        }
        }
        return -1;
}
int main(){
        int arr[]={2,3,4,5,8};
        int size=sizeof(arr)/sizeof(arr[0]);
        int target;

        cout<<"Enter a number to search: ";
        cin >>target;

        int result=binarySearch(arr, size, target);
        if(result !=1){
        cout<<"Element found at index: "<<result<<endl;
        }else{
        cout<<"Element not found."<<endl;
        }
        }
```

OUTPUT:

Output

```
/tmp/l6aPPdyoJ1.o
Enter a number to search: 10
Element found at index: -1


=== Code Execution Successful ===
```